H-7: Hodge Paper Structure & T-Number Audit

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1. Two Papers, Paired

Paper H1: “The Hodge Conjecture via Theta Correspondence on D_IV^5” - File: `notes/BST_Hodge_Proof.md` (1429 lines, v23) - Target: Annals of Mathematics / Compositio - Scope: Layer 1 Shimura proof. Proves Hodge for arithmetic quotients of D_IV^5 via Kudla-Millson theta correspondence. Layer 2 AC(0) reformulation. Extension routes explored; definitive treatment deferred to [H2].

Paper H2: “Ring Uniqueness and the Hodge Conjecture: Why D_IV^5 and Nothing Else” - File: `notes/BST_Hodge_Paper_H2_Ring` (234 lines, v0.1) - Target: Annals companion or Duke Math Journal - Scope: Ring uniqueness (T1780), cross-type exclusion (Toy 2120), exclusion lemmas (6 classes), Horikawa violation (Toy 2121), over-determination (T1779), KS extension, scope statement.

Why two papers (not one): Different audiences, different referees, different reading patterns. H1 is technical construction (theta correspondence, automorphic forms, Kudla-Millson — referees in automorphic forms). H2 is structural classification (bounded symmetric domains, Chern rings, exclusion — referees in algebraic geometry/Lie theory). Same reasoning as YM #76 vs #77.

Cross-references: H1 Section 1.3 Theorem 1.3 cites [H2] for ring uniqueness. H2 Section 1 cites [H1] for the construction proof. H1 Layer 3 explicitly defers to [H2]. Clean separation, no duplication.

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2. Paper H1 — Section Map with T-Numbers

Section	Title	T-numbers	Toys	Status
1.1	The Problem	—	—	Done
1.2	The Dictionary	T104	—	Done
1.3	Main Results	T114, T152, T421, T422, T1779	398, 399, 400, 401, 402, 2120	Done (v23)
1.4	Method	T104, T108, T110, T112, T113	—	Done
2	Background	—	—	Done
3	Layer 1: Theta Correspondence	T108, T110, T111, T112, T119, T149	398, 399, 401, 402, 1014, 1370	Done
3.x	Boundary cohomology	T124, T116, T115	401	Done
4	Layer 2: AC(0) Reformulation	T104, T108, T110, T112, T113, T114, T421, T422	400	Done
4.3	T104 Applied to Hodge	T104, T112	398, 399, 402	Done

Section	Title	T-numbers	Toys	Status
5	Layer 3: Extension Routes	T148, T149, T150, T151, T152, T153, T570, T600	1009, 1010, 1014	Done (defers to [H2])
5.7	Group-Independent Lift	T151	—	Done
5.8	Weight-Independent (T152)	T104, T152	—	Done
5.9	Two-Path Proof	T152, T153	—	Done
6	Confidence Assessment	—	All	Done (v23, D/I/C/S tiers)

H1 T-number inventory: 21 theorems referenced - Core proof chain: T108 -> T110 -> T112 -> T113 -> T114 - Spectral machinery: T111, T115, T116, T119, T124, T148, T149 - AC framework: T104, T147, T150, T151, T152, T153, T421, T422, T570, T600 - Companion ref: T1779

H1 Toy inventory: 398, 399, 400, 401, 402, 1009, 1010, 1014, 1370, 2120

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3. Paper H2 — Section Map with T-Numbers

Section	Title	T-numbers	Toys	Status
1	Introduction	T1779, T1780, T1781, T1782	—	Done
2	The Five Constraints	T1780	—	Done
2.1	Constraint 1: Diagonal Hodge + Kottwitz	—	—	Done
2.2	Constraint 2: Theta saturation	—	—	Done
2.3	Constraint 3: Selberg degree	—	—	Done
2.4	Constraint 4: Spectral gap + Wallach	—	1856, 2122	Done (Cal flag resolved)
2.5	Constraint 5: Hodge filtration	—	—	Done
3	Cross-Type Exclusion	T1781	2120	Done
4	Exclusion Lemmas	NEEDS T-NUMBERS	2120	Done (but see gap below)
5	Over-Determination	T1779	2119	Done
6	Kuga-Satake Extension	T1759	2097	Done
7	Discussion	T1743, T1756, T1782	2121	Done
8	Conclusion	—	—	Done

H2 T-number inventory: 7 theorems referenced (T1743, T1756, T1779, T1780, T1781, T1782, T1783)

H2 Toy inventory: 1399, 1856, 2119, 2120, 2121, 2122

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4. T-Number Coverage Audit

4.1 All referenced T-numbers (39 total) — all present in graph

TID	Name	In H1	In H2	In Graph
T104	Amplitude-Frequency Separation	Yes	—	Yes
T108	BMM H^{1,1} (Hodge, codim 1)	Yes	—	Yes
T110	B_2 Representation Filter	Yes	—	Yes
T111	Theta Lift Surjectivity (codim 1)	Yes	—	Yes
T112	Theta Lift Obstruction (codim 2)	Yes	—	Yes
T113	Phantom Hodge Exclusion	Yes	—	Yes
T114	Hodge Depth Reduction	Yes	—	Yes
T115	Tate Conjecture for SO(5,2) Shimura	Yes	—	Yes
T116	Absolute Hodge Classes on D_IV^5	Yes	—	Yes
T119	Lefschetz-Hodge for Type IV	Yes	—	Yes
T124	Eisenstein Controls Boundary Hodge	Yes	—	Yes
T147	BST-AC Structural Isomorphism	Yes	—	Yes
T148	Metaplectic Splitting Dichotomy	Yes	—	Yes
T149	Uniform Rallis Non-vanishing	Yes	—	Yes
T150	Induction Is Complete	Yes	—	Yes
T151	Group-Independent Lift	Yes	—	Yes
T152	Hodge = T104 on K_0	Yes	—	Yes

TID	Name	In H1	In H2	In Graph
T153	The Planck Condition	Yes	—	Yes
T421	Depth-1 Ceiling	Yes	—	Yes
T422	(C,D) Framework	Yes	—	Yes
T570	Hodge Linearization	Yes	—	Yes
T600	DPI (bridge)	Yes	—	Yes
T704	D_IV^5 Uniqueness Theorem	—	(implicit)	Yes
T832	Chern Class Integer Theorem	—	(ecosystem)	Yes
T834	Chern Rosetta Stone	—	(ecosystem)	Yes
T1000	Hodge CM Density	—	(ecosystem)	Yes
T1275	Hodge Physical-Uniqueness Closure	—	(ecosystem)	Yes
T1465	BSD CLOSED: Chern hole	—	(ecosystem)	Yes
T1667	Selberg Zero Moduli = Chern Classes	—	(ecosystem)	Yes
T1743	RH Discrimination (four filters)	—	Yes	Yes
T1756	BBW DOF Position (Conj. 3.2 RESOLVED)	—	Yes	Yes
T1757	Weight-2 Uniqueness + Pure Hodge Type	—	(ecosystem)	Yes
T1759	Hodge KS Frontier	—	Yes	Yes
T1760	T1269 Role Audit	—	(ecosystem)	Yes
T1761	D_IV^5 Universality	—	(ecosystem)	Yes
T1779	Hodge Over-Determination (H-3)	Yes	Yes	Yes
T1780	Hodge Ring Uniqueness (H-1)	—	Yes	Yes
T1781	Hodge Cross-Type Cascade (H-2)	—	Yes	Yes
T1782	Hodge Violation at Horikawa (H-4)	—	Yes	Yes

TID	Name	In H1	In H2	In Graph
T1783	Chern Sum Uniqueness (H-10)	—	Yes	Yes

4.2 Missing graph edges (7 needed)

The Hodge-ecosystem theorems have 28 internal edges and 162 cross-domain edges, but 7 expected connections are missing:

From	To	Reason
T108 (BMM H^{1,1})	T110 (B_2 Filter)	H1 proof chain: Step 1 feeds Step 2
T108 (BMM H^{1,1})	T113 (Phantom Exclusion)	H1: codim-1 result is input to phantom argument
T104 (Amplitude-Freq)	T113 (Phantom Exclusion)	H1 Section 4.3: T104 is the mechanism for phantom exclusion
T149 (Rallis Non-vanishing)	T112 (Theta Obstruction)	H1: Rallis certifies theta lift at codim 2
T119 (Lefschetz-Hodge)	T108 (BMM H^{1,1})	H1: Lefschetz is the codim-1 base case
T1779 (Over-determination)	T1780 (Ring Uniqueness)	H2: over-determination supports ring uniqueness
T152 (Hodge=T104 on K_0)	T1275 (Physical Closure)	Bridge: Layer 2 general proof -> closure theorem

Action: Grace should add these 7 edges to `play/ac_graph_data.json`. All are `derived` type.

4.3 T-number gap: Exclusion Lemmas need T-numbers

Paper H2 Section 4 contains four exclusion lemmas (4.1-4.4) that are currently unnamed in the theorem registry:

Lemma	Content	Proposed T-number
4.1	Non-orthogonal types excluded (15 domains)	Claim T1784
4.2	Even Type IV excluded (8 domains, Kottwitz + tube)	Claim T1785
4.3	IV_3 near-miss (m_s < 3, unitarity)	Claim T1786
4.4	Large odd Type IV (d_F >= 3, Rallis)	Claim T1787

These four lemmas are the formal backbone of the exclusion cascade (Toy 2120). They should be registered as theorems with edges to T1781 (cascade) and T1780 (ring uniqueness).

Action: Claim T1784-T1787 via `./play/claim_number.sh theorem` (4 claims). Register with edges: - T1784 -> T1781, T1785 -> T1781, T1786 -> T1781, T1787 -> T1781 (all feed the cascade) - T1780 -> T1784, T1780 -> T1785, T1780 -> T1786, T1780 -> T1787 (ring uniqueness implies each)

4.4 Connections to existing theorems

Connection	Via	Status
T1743 (RH four filters) -> T1780 (Hodge ring uniqueness)	Same F4+F6 squeeze applies to both RH and Hodge	Edge EXISTS
T704 (D_IV^5 Uniqueness) -> T1780	Ring uniqueness refines D_IV^5 uniqueness	Edge via T1780 -> T704: EXISTS
T1856 (Chern-beta dictionary)	NOT IN GRAPH	See note below
T1756 (BSD Conj. 3.2) -> T1779 (Over-determination)	BSD integers confirm Hodge integers	Edge EXISTS
T1275 (Physical Closure) -> T1780	Closure theorem updated by ring uniqueness	Edge EXISTS (bidirectional)
T1761 (D_IV^5 Universality) -> T1275	Universality subsumes closure	Edge EXISTS

Note on T1856: The Chern-beta dictionary is referenced as "Toy 1856" throughout, but it was listed in the CI_BOARD existing assets. It appears in graph as a toy reference on T1780 (toys field is empty — should be updated to include Toy 1856). This is a data-layer issue for Grace (H-9).

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5. Recommended Submission Structure

5.1 H1 + H2 paired submission

Submit together with cover letter explaining the pair: - H1: The construction. "We prove Hodge for D_IV^5 Shimura varieties." - H2: The uniqueness. "We prove this is the only domain where the proof works."

Same journal (Annals) or split (H1 Annals, H2 Inventiones/Duke). Casey decision.

5.2 Over-determination as standalone (Cal recommendation)

Extract T1779 material from H2 Section 5 as a 10-15 page Bulletin AMS perspective piece: "Five Integers, Thirty-Three Constraints: BST as Over-Determined System." Different audience (broad math community), different purpose (the convergence argument). Casey decision on timing.

5.3 Paper numbering

Current BST paper numbering goes through #103. Hodge papers should be: - Paper H1 = Paper #104 (or next available in queue — #104-#117 are "SE paper pipeline") - Paper H2 = Paper #105 - Over-determination perspective = unnumbered (outreach) or Paper #106

Casey should decide whether Hodge displaces SE pipeline or gets its own numbers.

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6. Completeness Checklist

Item	Status
All T-numbers in H1 present in graph	YES (21/21)
All T-numbers in H2 present in graph	YES (7/7)
All toys in H1 present as files	VERIFY (10 toys)
All toys in H2 present as files	VERIFY (6 toys)

Item	Status
H1 cross-references H2 correctly	YES (Layer 3 defers to [H2])
H2 cross-references H1 correctly	YES (Section 1, References)
T1779 text says "T1779" consistently	CHECK — H1 refs T1779 for ring uniqueness but T1780 is the ring uniqueness theorem. FIX NEEDED: H1 Layer 3 says "ring uniqueness, T1779" but should say "ring uniqueness, T1780"
Exclusion lemmas have T-numbers	NO — claim T1784-T1787
Missing graph edges added	NO — 7 edges needed
Toy 1856 linked to T1780 in graph	NO — update toys field
Cal flag 5 (triple coincidence source)	RESOLVED — Section 3.2 added to H2 v0.2: three Type IV identities + Toy 1399 T6 citation

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7. Actions Required

Immediate (before Phase 4 cold-read)

Action	Owner	Priority
Fix T1779/T1780 reference in H1: Layer 3 says "ring uniqueness, T1779" — should be T1780	Lyra	HIGH
Claim T1784-T1787: Four exclusion lemma theorems	Grace	HIGH
Add 7 missing edges to graph	Grace	HIGH
Add Toy 1856 to T1780 toys field in graph	Grace	MEDIUM
~~Verify Cal flag 5: Source of n_C^2 - 5n_C = 0 equation~~	Lyra	DONE
Paper numbering decision	Casey	LOW
Over-determination standalone decision	Casey	LOW

Already complete

Item	Done by
Paper H1 v23 finalized	Lyra
Paper H2 v0.1 written, all Cal flags addressed	Lyra
All 5 Cal Phase 3 flags resolved (except flag 5)	Lyra+Elie
Toy 2122 Chern sum independence verified	Elie
Exclusion lemmas written	Lyra

Revision History

- v1.0 (May 11, 2026): Initial H-7 deliverable. Two papers mapped, 39 T-numbers audited, 7 missing edges identified, 4 exclusion lemma T-numbers proposed (T1784-T1787), T1779/T1780 reference error caught.